

The Concept of Analogy

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Introduction

1. In this treatise I have set myself the task of continuing the closer investigation of some of the fundamental concepts which, in my book *Den menneskelige Tanke* [The Human Thought] (1910), I attempted to exhibit as the most characteristic for human cognition, such as we know it after its development hitherto. I have thus wished to give a companion piece to the treatise on *Totalitet som Kategori* (Videnskabernes Selskabs Skrifter. 1917) and to the treatise on *Relation som Kategori* (Videnskabernes Selskabs filosofiske Meddelelser. 1920), and it is the concept of analogy that I shall here investigate more closely.

Provisionally, analogy can be defined as likeness of relation between two subject-matters, that is to say, a likeness which does not hold of single properties of those subject-matters or of single parts of them, but of the relation of properties to one another or of parts to one another. There may be a very great difference between two subject-matters' properties or parts, each taken by itself, and yet there may be a likeness in the relation between the properties or between the parts. Analogy is thus a peculiar kind of likeness. It stands in a certain sense further from identity (absolute likeness) than qualitative likeness, composite likeness and coincidence do, which concern only single properties or parts, and which more easily form transitional forms to identity (cf. *Den menneskelige Tanke* § 21). Yet it must be remarked that there is not only an identity of subject-matter, but also an identity of relation, and already Aristotle distinguishes between quantitative analogy (ἰσότης λόγων), proportionality in the strict sense, and qualitative analogy, which may hold, e.g., between geometrical figures or between organs in different living beings. And there is besides an identity of difference, which lies at the base of the formation of an identically varying difference-series (*Den menneskelige Tanke* § 65) and especially of the formation of the identical quality-series that are fundamental

for exact science (number, time, degree, place) (*Den menneskelige Tanke* §§ 76–82).

Both totality and relation I have given a place among the categories, that is to say, among the fundamental concepts, the forms of thought, which at science's present stage play an essential role in thought-work, determining its tasks and its whole direction (*Den menneskelige Tanke* §§ 51–52; 57). Neither Aristotle nor Kant has given analogy a place among the categories, and in *Den menneskelige Tanke* I followed them in this. Yet in that work I was led to lay much weight on the fact that, in contrast to purely formal thinking (logic and arithmetic), analogy plays an essential role in all thinking about real subject-matters. The opposition between identity (identity of subject-matter in logic, identity of difference and identity of relation in arithmetic) and analogy is therefore of great importance for the right apprehension of the nature and the validity of our cognition. But when analogy really plays so great a role, while identity is properly an unattainable ideal outside the formal world of abstractions, it might be warranted to give it a place in the series of the categories; and in the table of categories to be found on the last page of the treatise *Totalitet som Kategori* this has in fact been done. It is entered there among the formal categories, immediately after the concept of identity, because it lies at the base of that reduction of relations of quality which — with the help of points of view of identity — takes place in the stricter development of the concepts of number, time, degree and place. Such series form intermediate members between absolute identity-series and qualitatively different series of subject-matters. They are formed by analogizing abstraction from the latter. It is here that analogy especially shows its importance. It should now be the task of the present treatise to follow the concept of analogy further in its more special importance for human cognition.

I dwell first on the involuntary analogies which make themselves felt in primitive men and in children, but also, at the high points of the life of the spirit, in the genius. They do not come forward as analogies for the consciousness in question itself; that is possible only for the critical consciousness, which has learned to distinguish between analogy and identity and to see that every grounding and proving transition of thought takes place in virtue of identity. Thereafter I go on to investigate analogy from a formal logical point of view, and in the following sections I seek to exhibit analogies between cognition's different functions and between its different domains.

Provisionally it may be remarked that analogy's importance for cognition shows itself

in three different respects.

Its greatest importance it certainly has by working in a scouting, anticipating and spurring manner. Categories in general lay themselves open to view in the questions that are put (*Den menneskelige Tanke* § 52). Analogy may be the actual road to true cognition, even if the analogy's validity is afterwards rejected. One can, after all, draw a correct conclusion from false premisses. Naturally one cannot prove that the conclusion is correct except by finding the right premisses; but it often happens that an assumption stands for us as true, and that precisely thereby it sets us the task of finding out why it is at bottom true. The premisses (the right premisses) are often the last thing to be found; but one would not seek after them if an analogy had not first led us to an assumption and thereby set us a task. The analytic method rests on this, that a task has been set us, and that one then seeks the conditions for solving that task. If the right premisses are then found, one perhaps rejects the analogy that showed the way — not only if it is false, but also if it is unnecessary. The ground for an analogy's being set up will lie in this, that there is a break in a continuous connection which has formed itself involuntarily for us, and which an involuntary need leads us to hold fast as long as possible. Analogy may here lead to an intellectual venture, which is sometimes crowned with success. One may apply to analogy what Pascal says of the imagination in general, that it deceives all the more easily because it does not always deceive (*d'autant plus fourbe qu'elle ne l'est pas toujours*). If it does not deceive, or if it shows itself to be dispensable, it is replaced by a determination of relation which is finally grounded in identity.

But analogy is not merely preparatory. It may be the last bond of union between different subject-matters or series of subject-matters which cannot be reduced in such a way that a single subject-matter or a single series of subject-matters is seen to lie at the base, or in such a way that they are all seen to have their ground in more deeply lying subject-matters or series of subject-matters. A certain rounding-off and survey becomes possible, in the face of such subject-matters, only by the help of analogy. We do not thereby obtain an explanation of why there are different subject-matters or series of subject-matters, but we establish the actual state of the case, namely that at certain points we get no further than to a connection, a synthesis, which builds only on the remotest degree of likeness — which analogy is. It will appear that both in the face of cognition's different functions (see ch. III) and in the face of its different domains (see ch. IV) we are

finally placed in this way. We end then in open questions, since it remains undecided whether a further reduction, a more deeply lying continuity, might be found, or whether there are here, in the nature of things, differences which are at any rate irreducible for us. Questions can thus be not only the beginning of wisdom, but also its end.

It will have its interest, finally, to dwell somewhat on analogies whose importance lies in their being expressions of poetic and religious moods — expressions which may recall now the analogies of involuntary mental life, now the seeking and finding of scientific thinking. Especially will intimations or hopes of a unity and a continuity lying deeper than experience can exhibit find expression in this kind of analogy.

Chapter I

Involuntary Analogies

2. As already remarked, analogy does not always presuppose that a distinction has expressly been drawn between different subject-matters, or respectively between their properties or parts, and that afterwards, in a certain opposition to the differences, a connection is brought about between the subject-matters on the basis of a likeness of relation. In involuntary mental phenomena, such as sense-intuition, memory and imagination, it cannot always be shown that the elements are first present as separated and only afterwards enter into combination. The importance of these involuntary mental processes lies precisely in this, that they bring about wholes, combinations, which can be analysed only afterwards. Reflection constantly needs a material, but this material need not, as has often been believed, consist in a multitude of independent elements; it may be wholes, which it is then reflection's task to resolve into elements that may not without more ado be regarded as identical with the elements to which the involuntarily formed wholes are due. For newly emerging elements may at once and involuntarily be woven together with consciousness's given content, so that perhaps no alteration at all is noticed. All the same, there is a tendency in reflection to regard the elements which its analysis finds as the whole's original, elementary foundation. There is here an analogy present, which at most can seek its justification in the continuity with which the transition can take place from an involuntarily formed mental continuum to the differentiation to which the very arising of reflection already testifies. Properly, when we speak of the most primitive and involuntary mental processes, all designations fall short, and this is no wonder, since language's psychological designations are fetched from the

clearly differentiating, critically reflecting mental life. The very expression “analogy” we here use in an analogizing manner. It is here as when one speaks of “unconscious inferences”. These are not inferences in the proper sense, but processes, transitions, which can be interpreted in analogy with the proper inferences. The same holds further (as I have already shown in my *Psykologi* and later in *Den menneskelige Tanke* § 5) when we set up the concept of synthesis as an expression for the nature of all conscious life. In epistemology, where we investigate the possibilities of scientific cognition, we must also discuss the possibility that the given might consist of a heap of equally great and mutually irreducible differences. Only an approximation to such a chaotic difference-series can be exhibited (as also to its counterpart, an absolute identity-series); there would in any case be needed a searching and strenuous reflection to establish that such a series really lay before us in a determinate given case.

In the manner in which a single lived experience, from the very moment of its conception, is connected with other simultaneous or past lived experiences, or with traditions into which the individual has lived himself so that these too can be regarded as lived experiences, there lies already an interpretation. The further back we come towards what we call the primitive man’s stage, the closer lived experience and interpretation move to one another, and finally there is no opposition between them to be found. When we nevertheless attempt to hold them apart, we make use of an analogy with what first makes itself felt for testing reflection. And, as has been said, the very concept of analogy is used analogically when we say that primitive men interpret every new lived experience in analogy with what they know from the traditions into which they have lived themselves. — In my treatise *Oplevelse og Tydning* I have sought to describe different ways in which the relation between lived experience and interpretation comes forward historically, as regards ecstatic experiences. —

In his important work *Les fonctions mentales dans les sociétés inférieures* (1910), LÉVY-BRUHL has criticized the formerly common conception that primitive man apprehends everything on the likeness of himself, anthropomorphizes or analogizes. It is, on his conception, not comparison or reflection that is at work here, but an entirely immediate identifying. In primitive man there always lie traditions at the base, with which the new that is lived through involuntarily melts together. The representations handed down from the tribe make themselves felt with such strength of feeling that no need

for reflection arises. What deviates in high degree from what has been handed down is provisionally not noticed; only later is an interpretation perhaps thought out that agrees with the tradition. When the savage sees an image of an object he knows, the image as image brings him nothing new, for the image is the object itself, however he may afterwards arrange this relation for himself. And with all the logical consistency which already distinguishes this stage — which Lévy-Bruhl has wrongly called the “prelogical” — the savage therefore fears to become dependent on one who has an image of him, and he fears that he will come to suffer want of game when a European has drawn or photographed many wild animals in the district and has departed with the pictures. And as it goes with images, so it goes also with names and with ceremonies: they are what they mean. Images, names and ceremonies are indeed only a part of a whole which has other properties besides those that are depicted or named; but this part *is* the whole. Lévy-Bruhl finds in this identifying of part and whole — what he calls “loi de participation”¹ — the most characteristic peculiarity of the primitive mental life. It is certain that the interpretation which primitive man institutes under the influence of a tradition embraced with heartfelt adherence and reverence cannot follow the principles of formal logic. These make themselves felt only when doubt has been awakened; but he is accustomed to regard the tradition as a self-evident foundation for his apprehension of everything that happens. The basis of interpretation is not the work of reflection. But it may lead to misunderstanding when Lévy-Bruhl declares (p. 425) that the collective representations governed by the “law of participation” are “indifferent to the logical law of contradiction”. A distinction must be drawn between the given basis of interpretation (here: the tradition held fast in reverence, which is once for all stamped by the “law of participation”) and the interpretation of single phenomena that takes place on the ground of this basis. Reflection works at all times and on all levels of culture on the ground of determinate presuppositions which cannot themselves at once be made an object for reflection. On the basis which primitive man once for all has, it is entirely consistent that he does not understand why the European distinguishes between the image and the thing. When Lévy-Bruhl calls “the primitive mentality” “prelogical”, this fits the basis of interpretation, not the interpretation made on the ground of that basis.² As there will

¹To this corresponds what GRØNBECH (*Primitiv Religion*, p. 8) calls “Helhedsrealisme” [realism of the whole], and which is characteristic for the primitive mental life.

²Dr. ALBERT SCHWEITZER, who works as a physician in the primeval forests of equatorial Africa, writes,

later be occasion to mention, Lévy-Bruhl has himself in another connection maintained, and rightly, that human consciousness, as regards the ultimate basis for judgments and inferences, properly never quite loses its “prelogical” character. There will always be presuppositions present which we do not and cannot bring forward, even in our most logical reflection. — As regards especially the close connection between name and thing, one can find examples of it even today. There is a tendency to regard the terminology one uses oneself as the only “right” one, and the language one speaks oneself seems to be one with the thing, while foreign languages’ designations often seem foreign to the things. It is told of a German who visited Paris that he reported home that there they called bread “pain”, and added, “Wir nennen es Brot, und es ist doch auch Brot!”

Tradition comes forward for the savage as a closed whole, within which he has no ground for making any distinction. There provisionally rules an immediate belief in the validity of the representations handed down, just as in the sense-perceptions that immediately appear. And the expression “belief in” is even too reflected, for no distinction at all is drawn between the belief and that which is believed in. The very concept of reality or validity is first formed when doubt has made itself felt. The fundamental proposition in accordance with which thinking provisionally goes (though it is of course not formulated) can be expressed by the question “Why not?” Why not rely on everything that comes forward for us with living force as sensation or as tradition? Only later does one learn to ask “Why?” — and reflection can then even reach the point of directing this question against the very thing that has hitherto formed the undisputed basis. For that it is required that new lived experiences should appear so numerous, and with such a peculiarity deviating from everything customary, that they can gain power over what one has already lived oneself into. ALBERT SCHWEITZER declares, as the result of several years’ living together with negroes in Central Africa: “One must not believe that one has got to the bottom of the negro’s world of thought if one has got no further than to look at his superstitious representations and his inherited rules of law. Innermost in him there lives a dark intimation that the apprehension of what is good must come by way of reflection” (*Mellem Floder og Urskove* p. 136). The self-sacrificing physician, who through historical criticism had learned to distinguish between the Christian dogmas and the

after several years’ stay among the negroes: “The natural man thinks more deeply than one commonly believes. Even if he can neither read nor write, he yet institutes reflections to a far wider extent than we suspect.” (*Mellem Floder og Urskov*. Danish transl. 1922. p. 134.)

ethical idealism that comes forward in the Sermon on the Mount and in the parables, experienced this especially when he sought to explain his religious apprehension to the negroes. Yet he experienced also that, alongside the “morality of the heart” that could be called forth in them, there long maintained itself a desire to lie and to steal, and a need for revenge which according to their own traditions was their right and perhaps even their duty.

I must here interpolate the remark that when in my *Psykologiske Undersøgelser* (p. 24 f.) I say that in the relation between new lived experiences and practised, customary representations the chief weight lies on the new, which “sets the whole process going”, this presupposes another stage of development than the undisturbed primitive mentality, where the new so easily glances off against the customary and is able to gain power in the mind only when it comes forward in a particularly insistent manner. —

If one asks after the origin of the system of collective representations that makes itself felt in the savage and determines his interpretation of what meets him, Lévy-Bruhl answers that it “is lost in the night of time”. And since every human group hitherto met with comes forward with certain collective representations, we cannot exhibit any absolutely first beginning. We must here, as with the question of the origin of language,³ content ourselves with analogies fetched from the levels of culture that lie nearest to us, and these analogies must of course be used with great caution. We see that on the levels of culture we know, the traditions that lie at the base of each new generation’s spiritual development undergo constant alterations, additions and displacements, which for an essential part are due precisely to the new generation’s own experiences and reflection. The collective is constantly altered under the influence of the individual. We cannot, as was often attempted in earlier times, think of a time when there was nothing collective, or when what collective there was was due exclusively to single individuals’ experience and reflection. But just as little can we think of a time when minds were governed exclusively by common traditions, without individual experiences and thoughts exercising any influence whatever. It must be assumed that on all levels of development there has taken place a reciprocal action between the common and the individual, and

³Cf. OTTO JESPERSEN: *Language, its Nature, Development and Origin*. (1922). p. 416 f. on the method in the discussion of the question of the origin of language. One must (as already, in their way, Madvig and Whitney had maintained) start out from the manner in which language actually develops, that is, from its history, and from there infer back to a more primitive language than any of those we know.

the difference of the levels can consist only in this, that on earlier levels the common, on later levels the individual, has the greatest influence — yet so that there is again and again a swinging to and fro between the different factors' predominance. Even on the most primitive levels we know, individual differences can make themselves felt and deviations from the tribe's belief come forward.⁴ And the examples of such things would undoubtedly be still more numerous if we had more knowledge of life within primitive societies than we have. We should have found remarkable deviations and initiatives which now are drowned for us in the ruling traditions.

Primitive conditions of life present great uniformity, and therewith great community, in the experiences which the single individuals can make. And on account of the constant living together in group and family, the single man's special experiences and inventions will very quickly become common property, and his name will be forgotten. Added to this, the practical demand constantly makes itself felt; action is required, and life is for a great part lived as reflex-life and instinct-life. Every representation gets its essential importance by releasing movement. But for that it is not necessary to enter into the whole connection to which the representation's content belongs. A single part or a single property is enough to release action, and one need not distinguish between part and whole; the part stands involuntarily as a sign of the whole. The image, the name or the sign works as if the object itself were given in it. The instinct has enough in an intimation to begin to stir. In the fact that a part can thus have the same influence as the whole, and that it may even be advantageous that time and force are not expended on investigating whether the whole is really represented by the single part or property, there might lie an access to an understanding of the "law of participation" or the "realism of the whole". And when a part thus shows itself again and again, in the race's experience, to have the same practical importance as the whole, one sees the possibility that there may be formed a circle of collective representations which obey the law of participation, and which it may be difficult for new lived experiences to break.

Purely logically, this identifying of part and whole (*pars pro toto*) can be formulated as an inference in the second figure: the part releases a certain movement, which the whole would release; therefore part and whole are the same! And this is precisely one of the ways in which an inference from analogy can be formulated. When an instinctive

⁴See my treatise *Sociologi og Religionsfilosofi* (in the book *Oplevelse og Tydning*. 1918) p. 131–138.

action can lead astray, this hangs together with the fact that the “inference” in question is wrong. Only within quite determinate limits, which the consciousness governed by an unconditional impulse to action does not give itself time to establish, can it do to identify part and whole. Neither instinct nor prelogical mentality can therefore maintain itself longer than the inference from analogy practically suffices. Doubt will be awakened when this is no longer the case. A closer psychological investigation will be able to elucidate this relation more nearly and thereby cast light on the involuntary analogies.

Chapter II

Analogy and Logic

Chapter III

Analogy between Functions of Cognition

Chapter IV

Analogy between Domains of Cognition

16. Logic and Arithmetic

17. Arithmetic and Geometry

18. Formal and Real Science

19. Natural Science and the Human Sciences

20. Ethics and Theoretical Science

21. World-view and Science

Chapter V

Poetic and Religious Symbolism